

ASSIGNMENT 06

Title : Demonstrate how to simulate a Data Center with 1 Host and run one cloudlet on it using Cloudsim.

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Code (SingleCloudletSimulation.java) :-

```
import java.util.*;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.*;

public class SingleCloudletSimulation {
    public static void main(String[] args) {
        try {
            CloudSim.init(1, Calendar.getInstance(), false);

            // Host
            List<Pe> peList = new ArrayList<Pe>();
            peList.add(new Pe(0, new PeProvisionerSimple(1000)));

            Host host = new Host(
                0,
                new RamProvisionerSimple(2048),
                new BwProvisionerSimple(10000),
                1000000,
                peList,
                new VmSchedulerTimeShared(peList)
            );

            List<Host> hostList = new ArrayList<Host>();
            hostList.add(host);

            // Datacenter
            DatacenterCharacteristics characteristics =
                new DatacenterCharacteristics(
                    "x86", "Linux", "Xen",
                    hostList, 10.0, 3.0, 0.05, 0.001, 0.0
                );

            Datacenter datacenter = new Datacenter(
                "Datacenter_0",
                characteristics,
                new VmAllocationPolicySimple(hostList),
                new LinkedList<Storage>(),
                0
            );

            // Broker
```

```

DatacenterBroker broker = new DatacenterBroker("Broker");
int brokerId = broker.getId();

// VM
Vm vm = new Vm(
    0, brokerId, 1000, 1, 512, 1000,
    10000, "Xen",
    new CloudletSchedulerTimeShared()
);

List<Vm> vmList = new ArrayList<Vm>();
vmList.add(vm);
broker.submitVmList(vmList);

// Cloudlet
Cloudlet cloudlet = new Cloudlet(
    0, 4000, 1, 300, 300,
    new UtilizationModelFull(),
    new UtilizationModelFull(),
    new UtilizationModelFull()
);

cloudlet.setUserId(brokerId);

List<Cloudlet> cloudletList = new ArrayList<Cloudlet>();
cloudletList.add(cloudlet);
broker.submitCloudletList(cloudletList);

// Simulation
CloudSim.startSimulation();
CloudSim.stopSimulation();

// Result
List<Cloudlet> resultList = broker.getCloudletReceivedList();
Cloudlet result = resultList.get(0);

System.out.println("Cloudlet ID: " + result.getCloudletId());
System.out.println("Status: SUCCESS");
System.out.println("Start Time: " + result.getExecStartTime());
System.out.println("Finish Time: " + result.getFinishTime());

} catch (Exception e) {
    e.printStackTrace();
}
}
}

```

OUTPUT :-

===== CLOUDSIM SIMULATION RESULT =====

Cloudlet Execution Summary

Cloudlet ID : 0
Execution Status : SUCCESS
Assigned VM : VM #0
Datacenter ID : Datacenter_0

Start Time : 0.0 sec
Finish Time : 4.0 sec
Execution Time : 4.0 sec

Observation:

- Cloudlet was successfully executed on the virtual machine.
- Total execution time = 4 seconds.

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Simulation completed successfully!

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